



Habib University
shaping futures

Instructions

- The exam will begin at **5:45 PM** and conclude at **7:15 PM**, with a total duration of **1 hour and 30 minutes**. The exam is worth **30 points** in total.
- This is a closed books/notes exam. Any notes, books, bags should be kept at the front of the classroom before the start of the test.
- Use of cellphones, laptops or any other communicating device is prohibited. Your cellphones should be switched off and submitted to the invigilator, or kept at the front of the class room before the start of the test.
- You are to attempt all question on the answer sheet provided. Your solutions should be comprehensible. Explain what you are doing, and if you use any results that you have studied in this course or Calculus 1 and 2, mention them explicitly.
- Use of calculators is allowed for this exam. Any form of communication/sharing among your peers is not allowed during the exam.
- If you want to communicate with the invigilator, raise your hand.
- You are not allowed to leave your desk without permission of the invigilator during the exam.
- Failure to abide by the instructions above would lead to immediate cancellation of the exam.

Student Name: _____

Instructor's Name: _____

Section: _____

1. (8 points) Given the vector field $\mathbf{F}(x, y, z) = z^2\mathbf{i} + xy\mathbf{j} + \sin z\mathbf{k}$.
 1. Calculate divergence of $\mathbf{F}(x, y, z)$.
 2. Describe whether the field is expanding, contracting or neither at these points.
 - (a) $P = (1, 2, 0)$
 - (b) $Q = (0, 1, \pi)$
 3. Calculate the curl of the vector field $\mathbf{F}(x, y, z)$.
 4. Let $\mathbf{F}(x, y, z) = F_1(x, y, z)\mathbf{i} + F_2(x, y, z)\mathbf{j} + F_3(x, y, z)\mathbf{k}$ and $\mathbf{G}(x, y, z) = G_1(x, y, z)\mathbf{i} + G_2(x, y, z)\mathbf{j} + G_3(x, y, z)\mathbf{k}$, be two vector fields in three dimensional space. Show that $\text{div}(\mathbf{F} + \mathbf{G}) = \text{div}(\mathbf{F}) + \text{div}(\mathbf{G})$.

Solution:**1. Divergence:**

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(z^2) + \frac{\partial}{\partial y}(xy) + \frac{\partial}{\partial z}(\sin z)$$

$$\nabla \cdot \mathbf{F} = 0 + x + \cos z = x + \cos z$$

2. At points:

(a) $P = (1, 2, 0)$,

$$\nabla \cdot \mathbf{F}(1, 2, 0) = 1 + \cos(0) = 1 + 1 = 2$$

The Field is expanding.

(b) $Q = (0, 1, \pi)$,

$$\nabla \cdot \mathbf{F}(0, 1, \pi) = 0 + \cos(\pi) = -1$$

The Field is contracting.

3. Curl:

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z^2 & xy & \sin z \end{vmatrix}$$

$$\nabla \times \mathbf{F} = (y\mathbf{k}) + (2z\mathbf{j}) = y\mathbf{k} + 2z\mathbf{j}$$

Proof: $\text{Div}(\mathbf{F} + \mathbf{G}) = \text{Div}(\mathbf{F}) + \text{Div}(\mathbf{G})$

Let $\mathbf{F}(x, y, z) = F_1(x, y, z)\mathbf{i} + F_2(x, y, z)\mathbf{j} + F_3(x, y, z)\mathbf{k}$ and $\mathbf{G}(x, y, z) = G_1(x, y, z)\mathbf{i} + G_2(x, y, z)\mathbf{j} + G_3(x, y, z)\mathbf{k}$ be two vector fields in three-dimensional space.

The divergence of a vector field $\mathbf{F}(x, y, z)$ is defined as:

$$\text{Div}(\mathbf{F}) = \nabla \cdot \mathbf{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

Similarly, the divergence of the vector field $\mathbf{G}(x, y, z)$ is:

$$\text{Div}(\mathbf{G}) = \nabla \cdot \mathbf{G} = \frac{\partial G_1}{\partial x} + \frac{\partial G_2}{\partial y} + \frac{\partial G_3}{\partial z}$$

Now, consider the sum of the two vector fields $\mathbf{F} + \mathbf{G}$:

$$\mathbf{F}(x, y, z) + \mathbf{G}(x, y, z) = (F_1 + G_1)\mathbf{i} + (F_2 + G_2)\mathbf{j} + (F_3 + G_3)\mathbf{k}$$

The divergence of the sum $\mathbf{F} + \mathbf{G}$ is:

$$\text{Div}(\mathbf{F} + \mathbf{G}) = \nabla \cdot (\mathbf{F} + \mathbf{G}) = \frac{\partial}{\partial x}(F_1 + G_1) + \frac{\partial}{\partial y}(F_2 + G_2) + \frac{\partial}{\partial z}(F_3 + G_3)$$

Using the linearity of partial derivatives, this becomes:

$$\text{Div}(\mathbf{F} + \mathbf{G}) = \left(\frac{\partial F_1}{\partial x} + \frac{\partial G_1}{\partial x} \right) + \left(\frac{\partial F_2}{\partial y} + \frac{\partial G_2}{\partial y} \right) + \left(\frac{\partial F_3}{\partial z} + \frac{\partial G_3}{\partial z} \right)$$

Grouping the terms together:

$$\text{Div}(\mathbf{F} + \mathbf{G}) = \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \right) + \left(\frac{\partial G_1}{\partial x} + \frac{\partial G_2}{\partial y} + \frac{\partial G_3}{\partial z} \right)$$

This simplifies to:

$$\text{Div}(\mathbf{F} + \mathbf{G}) = \text{Div}(\mathbf{F}) + \text{Div}(\mathbf{G})$$

Thus, we have proven that:

$$\boxed{\text{Div}(\mathbf{F} + \mathbf{G}) = \text{Div}(\mathbf{F}) + \text{Div}(\mathbf{G})}$$

2. (4 points) Let $\mathbf{F}(x, y, z) = x^2\mathbf{i} + yz\mathbf{j} + 2xz\mathbf{k}$ be a force field. Compute the work done by this force field in moving a particle along the straight line from $(0, 0, 0)$ to $(2, 3, 4)$.

Solution:

Let $\mathbf{F}(x, y, z) = x^2\mathbf{i} + yz\mathbf{j} + 2xz\mathbf{k}$.

Step 1: Parametrize the Path

The straight line from $(0, 0, 0)$ to $(2, 3, 4)$ can be parametrized as:

$$\mathbf{r}(t) = \langle 2t, 3t, 4t \rangle, \quad t \in [0, 1].$$

Step 2: Compute $\frac{d\mathbf{r}}{dt}$

$$\frac{d\mathbf{r}}{dt} = \langle 2, 3, 4 \rangle.$$

Step 3: Substitute into the Force Field

Substituting $\mathbf{r}(t)$ into $\mathbf{F}$:

$$\mathbf{F}(\mathbf{r}(t)) = (2t)^2\mathbf{i} + (3t)(4t)\mathbf{j} + 2(2t)(4t)\mathbf{k} = (4t^2, 12t^2, 16t^2).$$

Step 4: Compute the Work Done

The work done W is:

$$W = \int_0^1 \mathbf{F}(\mathbf{r}(t)) \cdot \frac{d\mathbf{r}}{dt} dt.$$

Calculating the dot product:

$$\mathbf{F}(\mathbf{r}(t)) \cdot \frac{d\mathbf{r}}{dt} = (4t^2, 12t^2, 16t^2) \cdot (2, 3, 4) = 8t^2 + 36t^2 + 64t^2 = 108t^2.$$

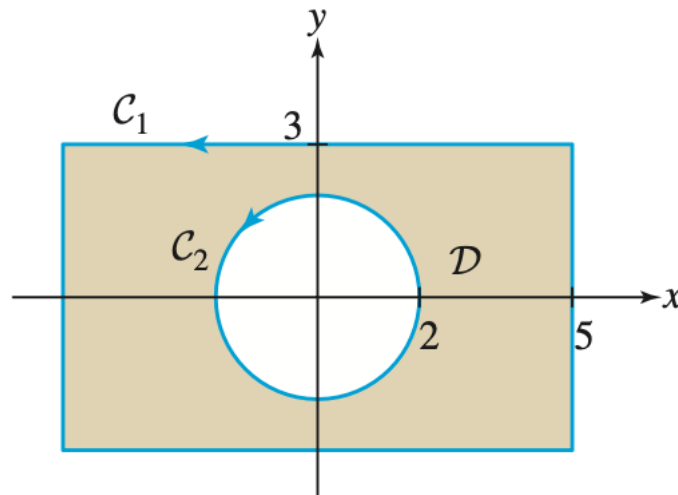
Thus, the work done is:

$$W = \int_0^1 108t^2 dt = 108 \left[\frac{t^3}{3} \right]_0^1 = 108 \cdot \frac{1}{3} = 36.$$

The work done by the force field is:

$$\boxed{36}.$$

3. (5 points) Referring to the figure below:



Suppose that $\oint_{C_2} \mathbf{F} \cdot d\mathbf{r} = 12$. Use Green's Theorem to determine $\oint_{C_1} \mathbf{F} \cdot d\mathbf{r}$, assuming that $\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} = -3$ in $\mathcal{D}$.

Solution: We are given that C_1 is the outer boundary (a rectangle), C_2 is the inner boundary (a circle), and the region between them is $\mathcal{D}$. The rectangle has length 10 and height 6, and the circle has radius 2.

From Green's Theorem, we have:

$$\oint_{C_1} \mathbf{F} \cdot d\mathbf{r} - \oint_{C_2} \mathbf{F} \cdot d\mathbf{r} = \iint_{\mathcal{D}} \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dA.$$

We are given:

$$\oint_{C_2} \mathbf{F} \cdot d\mathbf{r} = 12,$$

and

$$\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} = -3.$$

Thus, Green's Theorem becomes:

$$\oint_{C_1} \mathbf{F} \cdot d\mathbf{r} - 12 = -3 \cdot \text{Area}(\mathcal{D}).$$

Step 1: Area of Region $\mathcal{D}$

The area of the rectangle is:

$$\text{Area of the rectangle} = 10 \times 6 = 60.$$

The area of the circle (with radius $r = 2$) is:

$$\text{Area of the circle} = \pi r^2 = \pi \times 2^2 = 4\pi.$$

Thus, the area of region $\mathcal{D}$ is:

$$\text{Area}(\mathcal{D}) = 60 - 4\pi.$$

Step 2: Applying Green's Theorem

Now, substitute this area into Green's Theorem:

$$\oint_{C_1} \mathbf{F} \cdot d\mathbf{r} - 12 = -3 \cdot (60 - 4\pi).$$

Simplifying:

$$\oint_{C_1} \mathbf{F} \cdot d\mathbf{r} - 12 = -180 + 12\pi.$$

Thus:

$$\oint_{C_1} \mathbf{F} \cdot d\mathbf{r} = 12\pi - 168.$$

Step 3: Final Answer

The value of the line integral around C_1 is:

$$\boxed{\oint_{C_1} \mathbf{F} \cdot d\mathbf{r} = 12\pi - 168.}$$

4. (5 points) Given the vector field:

$$\mathbf{F}(x, y, z) = e^{x^2+y^2+z^2} (x \mathbf{i} + y \mathbf{j} + z \mathbf{k}),$$

1. Check if $\mathbf{F}$ is path independent. Clearly mention any theorems or results that you use for your conclusion.
2. Find the line integral of $\mathbf{F}$ along any curve from $(0, 0, \pi)$ to $(2, \frac{1}{2}, 0)$.

Solution:

1. Since $\mathbf{F}$ is continuous over $\mathbb{R}^3$ (three-dimensional space) and $\mathbb{R}^3$ is simply connected, by the Fundamental Theorem of Vector Calculus, $\mathbf{F}$ is path independent if and only if it is a gradient field. (1 mark for mentioning the Fundamental Theorem, 0.5 for also mentioning that $\mathbf{F}$ meets the conditions to apply the theorem)

Thus, we can check if $\mathbf{F}$ is a gradient field or not.

Test for gradient fields: $\mathbf{F} = P \hat{i} + Q \hat{j} + R \hat{k}$ is a gradient field if and only if

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

and

$$\frac{\partial P}{\partial z} = \frac{\partial R}{\partial x}$$

and

$$\frac{\partial Q}{\partial z} = \frac{\partial R}{\partial y}$$

For our field, $P = xe^{x^2+y^2+z^2}$, $Q = ye^{x^2+y^2+z^2}$, $R = ze^{x^2+y^2+z^2}$, we calculate:

$$\frac{\partial P}{\partial y} = 2xye^{x^2+y^2+z^2} = \frac{\partial Q}{\partial x}$$

and

$$\frac{\partial P}{\partial z} = 2xze^{x^2+y^2+z^2} = \frac{\partial R}{\partial x}$$

and

$$\frac{\partial Q}{\partial z} = 2yze^{x^2+y^2+z^2} = \frac{\partial R}{\partial y}$$

(1 mark for correctly checking the three equations)

Thus, the field $\mathbf{F}$ is a gradient field and, by the Fundamental Theorem, also path independent. (0.5 marks)

2. Since $\mathbf{F}$ is path independent by part (a),

$$\int_{\mathbf{C}}^{(2, \frac{1}{2}, 0)} \mathbf{F} \cdot d\mathbf{r} = f\left(2, \frac{1}{2}, 0\right) - f(0, 0, \pi)$$

where f is any potential function for $\mathbf{F}$.

Let's find f :

$$\begin{aligned} f(x, y, z) &= \int P dx + g(y, z) \\ &= \int x e^{x^2+y^2+z^2} dx + g(y, z) \end{aligned}$$

By substituting $u = x^2 + y^2 + z^2$ and $du = 2x dx$, the integral becomes:

$$\begin{aligned} &= \frac{1}{2} \int e^u du + g(y, z) \\ f(x, y, z) &= \frac{1}{2} e^{x^2+y^2+z^2} + g(y, z) \end{aligned}$$

We still need to identify $g(y, z)$ here. (1 mark)

Now, $\frac{\partial f}{\partial y}$ must match Q , thus:

$$\begin{aligned} \frac{\partial f}{\partial y} &= y e^{x^2+y^2+z^2} + \frac{\partial g}{\partial y} = y e^{x^2+y^2+z^2} = Q \\ &\Rightarrow \frac{\partial g}{\partial y} = 0 \end{aligned}$$

Thus, g is not dependent on y :

$$\Rightarrow g(y, z) = g(z)$$

Thus,

$$f(x, y, z) = \frac{1}{2} e^{x^2+y^2+z^2} + g(z)$$

Similarly, $\frac{\partial f}{\partial z}$ must match R , thus:

$$\begin{aligned} \frac{\partial f}{\partial z} &= z e^{x^2+y^2+z^2} + g'(z) = z e^{x^2+y^2+z^2} = R \\ &\Rightarrow g'(z) = 0 \end{aligned}$$

And so g is just a constant. We finally have:

$$f(x, y, z) = \frac{1}{2}e^{x^2+y^2+z^2} + C$$

(0.5 marks)

The line integral is:

$$\int_C^{(2, \frac{1}{2}, 0)} \mathbf{F} \cdot d\mathbf{r} = f\left(2, \frac{1}{2}, 0\right) - f(0, 0, \pi) = \frac{1}{2}\left(e^{17/4} - e^{\pi^2}\right)$$

(0.5 marks)

(Full marks if someone has correctly computed the line integral without using the Fundamental Theorem.)

5. (8 points) For each of the statements below, indicate whether they are **TRUE** or **FALSE**. In each case, provide a brief explanation or a counterexample for your choice.

1. The set of points (x, y, z) in 3-dimensional space such that $x^2 + y^2 = 4$ form a circle of radius 2.
2. A field with no source or sink has zero divergence everywhere.
3. The curl is nonzero if and only if the direction of the field changes.
4. If $\mathbf{F}$ and $\mathbf{G}$ are vector fields such that $\nabla \times \mathbf{F} = \nabla \times \mathbf{G}$, then $\mathbf{F} = \mathbf{G}$.

Solution: 0.5 marks for the correct choice (TRUE/FALSE), 1.5 for correct explanation

1. The set of points (x, y, z) in 3-dimensional space such that $x^2 + y^2 = 4$ form a circle of radius 2.
False. It forms an infinite cylinder centered at the origin, parallel to the z -axis, with radius 2.
2. A field with no source or sink has zero divergence everywhere.
True. By definition, a point in the domain of a vector field is called a source if there is positive divergence at that point. Similarly, a point in the domain of a vector field is called a sink if there is negative divergence at that point. Thus, a field with zero divergence everywhere has no source or sink in its domain.

3. The curl is nonzero if and only if the direction of the field changes.
False. $\mathbf{F} = x\hat{i} + y\hat{j}$ (the radial field in 2D) changes direction but has zero curl. Similarly, $\mathbf{F} = y\hat{i}$ doesn't change direction, but has $\text{curl} = 1$ everywhere.
4. If $\mathbf{F}$ and $\mathbf{G}$ are vector fields such that $\nabla \times \mathbf{F} = \nabla \times \mathbf{G}$, then $\mathbf{F} = \mathbf{G}$.
False. Many counterexamples can be given. For instance, $\mathbf{F} = y\hat{i}$ and $\mathbf{G} = -x\hat{j}$ both have $\text{curl} = 1$.

Important Formulae

- Assume sufficient differentiability of the functions and that the conditions for curves and surfaces are met, then:

Del Operator $\vec{\nabla}$	$\vec{\nabla} = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$
Gradient of a scalar function $\phi = \phi(x, y, z)$	$\vec{\nabla} \phi$
Divergence of a vector field $\vec{F}$	$\vec{\nabla} \cdot \vec{F}$
Curl of a vector field $\vec{F}$	$\vec{\nabla} \times \vec{F}$
Line integral of a vector field $\vec{F}$	$\int_C \vec{F} \cdot d\vec{r}$
Closed line integral	$\oint_C \vec{F} \cdot d\vec{r}$

- For conservative field $\vec{F} = F_x \hat{i} + F_y \hat{j} + F_z \hat{k}$,
 - For two different curves C and C' between points (x_a, y_a, z_a) and (x_b, y_b, z_b) ,

$$\int_{(x_a, y_a, z_a)_C}^{(x_b, y_b, z_b)} \vec{F} \cdot d\vec{r} = \int_{(x_a, y_a, z_a)_{C'}}^{(x_b, y_b, z_b)} \vec{F} \cdot d\vec{r}$$

- Over a closed curve C,

$$\oint_C \vec{F} \cdot d\vec{r} = 0$$

- There exists a scalar potential function f ,

$$\vec{F} = \vec{\nabla} f$$

- Test,

$$\frac{\partial F_x}{\partial y} = \frac{\partial F_y}{\partial x}, \frac{\partial F_x}{\partial z} = \frac{\partial F_z}{\partial x}, \frac{\partial F_y}{\partial z} = \frac{\partial F_z}{\partial y}$$

- Fundamental theorem of line integral,

$$\int_{(x_a, y_a, z_a)_C}^{(x_b, y_b, z_b)} \vec{F} \cdot d\vec{r} = \int_{(x_a, y_a, z_a)_C}^{(x_b, y_b, z_b)} \vec{\nabla} f \cdot d\vec{r} = f(x_b, y_b, z_b) - f(x_a, y_a, z_a)$$

- Green's theorem for a vector field $\vec{F} = F_x \hat{i} + F_y \hat{j}$,

$$\iint_S \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) dx dy = \oint_C \vec{F} \cdot d\vec{r}$$

Here the closed curve C is piecewise smooth, simple positively oriented and bounds the surface S in a two-dimensional xy-plane.